Glass-Silicon Heterogeneous Integration for System-in-Package of MEMS Differential Pressure Sensors

Zong Liu*, Chuncheng Che*, Xiaojuan Wu*, Jiaxing Wang*, Weilong Guo*, Jie Sun*, Taonan Zhang*, Nana He*, Xuecheng Hou*, Yifeng Qin*, Dongming Xing*, Qiuxu Wei*

*** Beijing BOE Sensor Technology Company, Ltd., Beijing, China**

Abstract

System-in-package integration of Micro-Electro-Mechanical Systems (MEMS) piezoresistive pressure sensors with Application-Specific Integrated Circuits (ASICs) is achieved using a glass interposer. Electrical interconnection via Through-Glass Vias (TGV) reduces package size versus conventional packaging. Wafer-level anodic bonding enables Si-glass heterogeneous integration. The interposer featuring circular bonding cavities and pressure-guiding channels for pressure equalization. Thus, it provides mechanical protection, electrical connection and pressure conduction, offering a novel approach for advanced system-in-package of miniaturized MEMS differential pressure sensors.

Author Keywords

Through Glass Via, MEMS, System-in-Package, Glass Interposer

1. Introduction

In the post-Moore era, heterogeneous integration has emerged as a paramount strategy to overcome the scaling limitations inherent in traditional integrated circuits. Among various integration paradigms, 2.5D/3D System-in-Package (SiP) technology distinguishes itself by enabling the simultaneous miniaturization and functional consolidation of multiple discrete chips.[1]

For MEMS differential pressure sensors, packaging technology is a critical determinant of measurement accuracy and long-term reliability. Conventional interposer materials, however, present significant challenges. Organic substrates are susceptible to thermal warpage and suffer from Coefficient of Thermal Expansion (CTE) mismatch, while silicon interposers are often constrained by high manufacturing costs and dielectric losses.[2] In contrast, glass interposers offer a compelling alternative, characterized by low dielectric loss, excellent thermal compatibility with silicon, and atomically flat surfaces, rendering them highly suitable for MEMS wafer-level bonding.[3]

Nevertheless, traditional glass-interposer packaging architectures impose limitations on the differential sensing capabilities of MEMS devices. Specifically, the hermetically sealed vacuum cavity typically formed via anodic bonding restricts the silicon diaphragm to single-sided pressure exposure, limiting the sensor to absolute pressure measurements.[4, 5]

To address this limitation, this study proposes a wafer-level glass-silicon heterogeneous integrated SiP architecture tailored for MEMS differential pressure sensors. The objectives of this work are twofold: (1) to preserve the heterogeneous integration advantages of glass interposers while enabling true differential pressure measurement; and (2) to achieve the monolithic integration of silicon MEMS, glass interposers, and ASICs through Laser-Induced Deep Etching (LIDE) TGV technology and anodic bonding, resulting in a miniaturized, cost-effective packaging solution.

2. Principle and Design

(a) Heterogeneous Integration Structure

The proposed design employs a three-layer vertically stacked heterogeneous architecture comprising MEMS, glass interposer, and ASIC layers, as illustrated in Figure 1(a). This configuration facilitates high-density system integration via wafer-level bonding, TGVs, and Redistribution Layer (RDL) routing. The intermediate layer consists of a borosilicate glass interposer, which serves as both a mechanical support structure and an electrical interconnection hub. The bottom surface of the interposer forms a heterogeneous bond with the silicon MEMS pressure sensor through anodic bonding. Vertical electrical conduction between the top and bottom surfaces is achieved via four 50 μm (bottom diameter) TGVs, fabricated using LIDE and precisely aligned with the heavily doped contact regions of the MEMS chip. An RDL on the top surface facilitates the redistribution of electrical interconnects for the ASIC chip and peripheral components. As depicted in Figure 1(b), piezoresistive signals from the MEMS sensor are transmitted to the ASIC through the TGVs, where they undergo amplification, temperature compensation, and analog-to-digital (A/D) conversion before outputting a standard current signal. This integration scheme reduces the signal path length to approximately one-fifth of that in conventional wire-bonding packages, effectively minimizing parasitic capacitance and transmission latency.

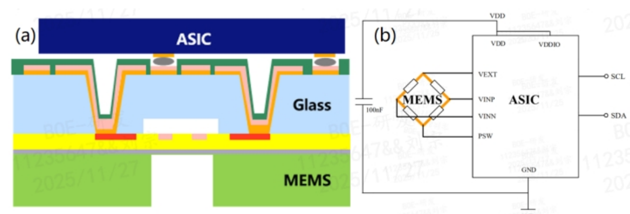

Figure 1. (a) Cross-sectional schematic of the SiP architecture for MEMS differential pressure sensors. (b) Schematic of internal electrical interconnections within the SiP.

(b) MEMS Piezoresistive Pressure Sensor

The operation of the silicon-based MEMS piezoresistive pressure sensor is governed by the piezoresistive effect of single-crystal silicon. Upon application of external pressure, the diaphragm undergoes bending deformation, inducing mechanical strain that modulates the resistivity of embedded piezoresistors. This modulation arises from lattice distortion, which alters charge carrier mobility within the silicon crystal lattice. Notably, the piezoresistive coefficients of single-crystal silicon exhibit pronounced anisotropy, with magnitudes varying significantly across different crystallographic orientations.

In this study, a Silicon-On-Insulator (SOI) wafer is utilized for sensor fabrication. Following etching, the device layer and the buried oxide layer collectively form the pressure-sensing

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diaphragm. Piezoresistive regions are defined via photolithography, followed by boron ion implantation to a depth of 1 µm to create P-type resistors. Four such doped regions are configured in a Wheatstone bridge arrangement (Figure 1(b)), converting resistance variations into measurable voltage outputs. The adjacent arms of the bridge are designed to experience opposing resistance changes, thereby enhancing sensitivity through differential signal processing. To maximize sensitivity, all piezoresistors fabricated on N-type <100> silicon wafers are aligned along the [110] crystal direction, leveraging the high piezoresistive coefficients characteristic of this orientation.

Finite Element Method (FEM) simulations were employed to determine the optimal diaphragm dimensions by analyzing stress distribution and deflection profiles under various pressure loads. Figure 3 presents the FEM simulation results for a diaphragm (614 × 614 × 9 µm) under 40 kPa pressure. As shown in Figure 2 (a), the maximum (1,2) stress component reaches 30.8 MPa, while Figure 2 (b) demonstrates a center deflection of the diaphragm of 1.05 µm. Stress analysis confirms that positioning piezoresistors at the mid-points of the diaphragm edges enables optimal strain capture.

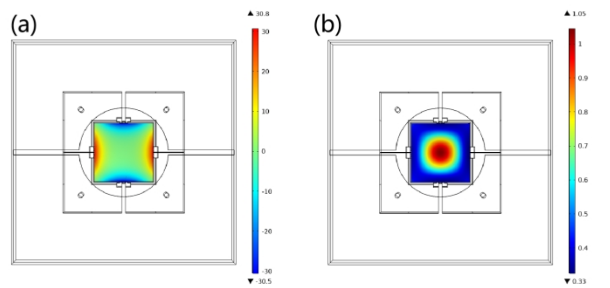

Figure 2. FEM simulation results for the diaphragm under 40 kPa pressure: (a) stress distribution of the (1,2) component on the top surface and (b) diaphragm displacement.

(c) Design of the Glass Interposer

The glass interposer in this study incorporates several key structural features including the channels and cavity after integration with silicon for differential pressure sensing as shown in Figure 3 (a).

Unlike conventional absolute pressure sensors that rely on unilateral pressure acting against a vacuum reference cavity, this design enables the independent conduction of dual pressure sources through an innovative architecture. The measured pressure (P_1) acts directly on the backside of the diaphragm via a silicon back cavity. Meanwhile, the reference pressure (P_2) enters a circular bonding cavity through an in-line pressure-guiding channel in the glass interposer and acts on the frontside of the diaphragm. The cross-section of the channels is arc-shaped, extending from the cavity edge to the interposer sidewall. This configuration ensures that the deformation of the diaphragm is determined by $\Delta P = P_1 - P_2$. The bonding cavity within the glass interposer is circular in shape, with a diameter of 900 µm, which fully covers the sensing diaphragm area. The channels with two in-line parallel paths flanking the circular cavity and collinear with the cavity center, forming redundant pressure conduction paths. This design maintains pressure conduction efficiency if one channel is blocked by particles, thereby reducing the probability of single - point failure.

The cavity depth is set at 50 µm and is fabricated in LIDE step, which provides sufficient displacement space for the diaphragm. A 150 µm spacing is maintained between the cavity edge and the piezoresistor region. This spacing is crucial as it reduce the interference of bonding stress with the piezoresistive properties of the sensors while maintaining the small size of the device. Finite element simulation was conducted to analyze the bonded interface stress resulting from the CTE mismatch after cooling to room temperature. The circular cavity in this design shows improved uniformity in stress distribution. After bonding at 300°C, when the structure returns to room temperature, the circumferential stress of the circular cavity is reduced to 13.6 MPa, as depicted in Figure 3 (b). This reduction in stress helps enhance the mechanical reliability of the overall device.

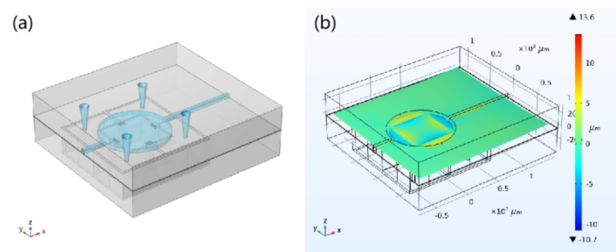

Figure 3. (a) 3D model of the glass-silicon heterogeneous integration. (b) Simulation result of the stress distribution.

3. Methods

The fabrication process adopted a comprehensive wafer-level workflow, which encompassed four core stages: silicon MEMS processing, glass interposer fabrication, heterogeneous bonding, and system integration, as illustrated in Figure 4.

An 8-inch SOI wafer, featuring a 0.5 µm buried oxide (BOX) layer and a 9 µm device layer, was processed as follows (Fig. 4a). Initially, the wafer underwent RCA standard cleaning. Subsequently, implantation regions were defined through photolithography. Boron ion implantation was carried out to form heavily-doped P-type conductive regions and P - type piezoresistors, followed by activation annealing (Fig. 4b).

An 8-inch borosilicate glass substrate was employed for the fabrication of the interposer (Fig. 4c). During this process, circular bonding cavities with a diameter of 900 µm and a depth of 50 µm, along with in-line pressure-guiding channels having a width of 50 µm and a depth of 50 µm, were patterned in a same LIDE step. After TGV arrays were patterned by another LIDE step, glass cavities and channels along with TGVs of an aspect ratio of 6:1 were fabricated by hydrofluoric (HF) wet etching (Fig. 4d). The cross-sections of the cavity and the channels are presented in Figure 5 (a) and (b), respectively.

The glass substrate was then thinned to 330 µm through mechanical grinding and chemical mechanical polishing (CMP) (Fig. 4d). This step ensured that the total thickness variation (TTV) across the 8-inch wafer was less than 1 µm. After CMP, the cross - section of the TGV is shown in Figure 5(c), while the top view of the structures, including TGVs, a cavity, and channels, is displayed in Figure 5(d).

Anodic bonding was utilized to heterogeneously integrate the silicon MEMS and the glass interposer, with the bonded interface area accounting for more than 85% of the total chip area (Fig. 4e). Anodic bonding is an electrochemical process that enables intimate bonding between silicon and glass wafers, driven by ion migration under an external electric field. In this process, the silicon wafer is biased as the anode and the glass as the cathode. When a direct-current (DC) voltage is applied. The SOI wafer was precisely aligned with the glass interposer. The bonding process was carried out in a vacuum environment with controlled mechanical pressure to prevent gas entrapment.

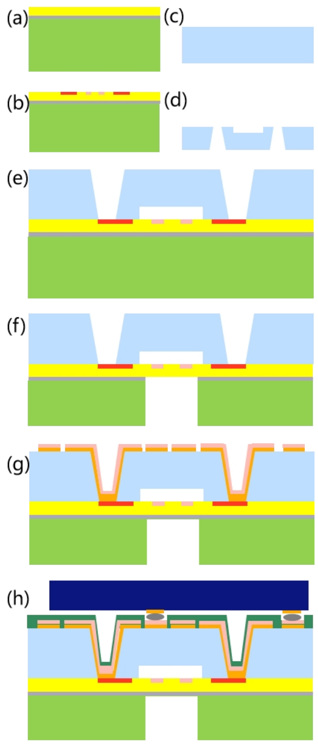

Figure 4. Cross-sectional schematic diagram of fabrication process flow

Following the bonding process, the silicon handle layer was thinned down to 400 µm through a combination of grinding and chemical - mechanical polishing (CMP), which results in an overall thickness of 730 µm. This thinning operation led to a over 40% reduction in the subsequent back - cavity etching time. Deep Reactive Ion Etching (DRIE) was subsequently carried out on the thinned silicon side to fabricate the sensing diaphragm. Specifically, a Bosch process, which involved alternating SF_6/C_4F_8 plasma cycles, was employed to etch the back cavity. The etching process was precisely halted at the BOX layer, resulting in a suspended silicon membrane that comprised the BOX layer and the device layer (Fig. 4f). Next, a titanium (Ti) adhesion layer and a copper (Cu) seed layer were sputtered onto the surface of the glass interposer. Electroplating was then used to increase the thickness of the Cu layer to 10 µm. After that, photolithography and wet etching techniques were applied to pattern the RDL. The topview of a chip after the formation of the RDL is shown in Figure 6(a). To prevent the copper on the surface from oxidizing or corroding and to enhance ohmic contact, an electroless nickel/gold (Ni/Au) layer was deposited (Fig. 4g). A polyimide (PI) solder mask was spin - coated onto the surface, and openings were defined at the pad locations via photolithography. The ASIC wafer was processed with a compatible RDL and tin (Sn) bumps to align with the pads on the glass interposer, facilitating heterogeneous integration. The ASIC dies were singulated from the wafer through dicing and then flip - chip bonded to the glass interposer using surface - mount technology (SMT) bumping (Fig. 4h). The top - view of a chip after ASIC integration is presented in Figure 6 (b). The silicon-glass-ASIC composite wafer was finally diced into individual chips. After dicing, individual chips with dimensions of 2 mm × 2 mm were obtained, with a dimensional deviation of less than 20 µm between the silicon and glass layers, ensuring high alignment precision for subsequent packaging.

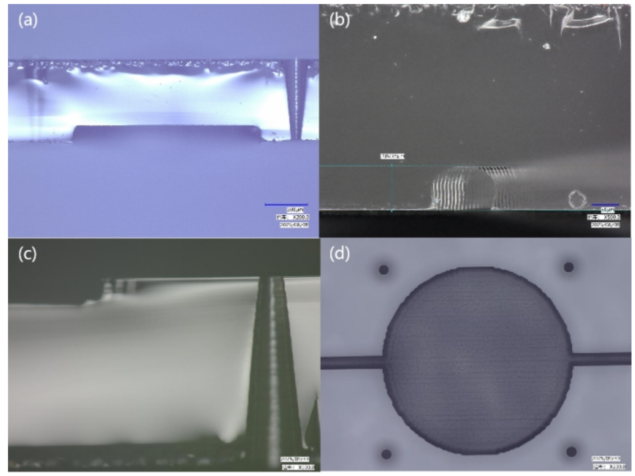

Figure 5. Cross-section of (a) the cavity and (b) the channels before CMP. (c) Cross-section of TGVs after CMP. (d) Top view of the structures including TGVs, a cavity and channels.

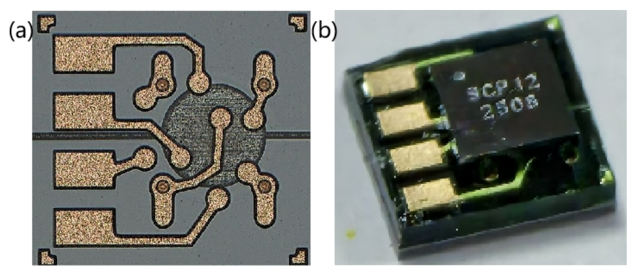

Figure 6. (a) Top view of a chip after formation of RDL. (b) A image of the complete device after integrating ASIC.

4. Results

The core performance parameters of the heterogeneously integrated MEMS differential pressure sensor developed in this study were characterized with a differential pressure measurement range of 0~40 kPa. The full-scale output signal reaches 100 mV. The linearity error was controlled to ±0.2% FS through least-squares fitting, while the hysteresis was minimized to ±0.1% FS. These values represent significant improvements compared to conventional organic substrates packaging solutions. The zero-offset temperature drift was measured less than 0.1% FS/°C across the temperature range from 0 to 65 °C. After integrated with the ASIC, while the the device in this study maintains comparable core performance metrics to the Sensata 112CP as outlined in Table 1, it exhibits superior electrical robustness with a wider overvoltage and reverse polarity protection range and enhanced mechanical durability via higher overload and burst test ratings. Collectively, these comprehensive performance metrics demonstrate the feasibility of glass-silicon heterogeneous integration for SiP of MEMS differential pressure sensors, particularly in terms of sensitivity, linearity, temperature stability, and environmental robustness.

Table 1. Device Performance Comparison

	Test Item	Unit	Sensata 112CP			Proposed Device		
			Min	Typ.	Max	Min	Typ.	Max
1	Supply Voltage	V	4.5	5	5.5	4.75	5	5.25
2	Output Voltage	V	0.5	-	4.5	0.5	-	4.5
3	Output Current	mA	8			-	-	6
4	Operating Temperature	℃	-40	-	130	-40	-	125
5	Storage Temperature	℃	-40	-	135	-40	-	130
6	Overvoltage & Reverse Polarity Protection	V	-14	-	16	-24	-	28
7	Overload Test	-	2X			3X		
8	Burst Test	-	3X			5X		

5. Discussion

This study presents a synergistic design that integrates a circular bonding cavity with bilaterally symmetric "in-line" pressure-guiding channels. The circular cavity allows for a uniform circumferential stress distribution. Compared to square cavities, it reduces the maximum stress on the silicon diaphragm and eliminates corner stress concentration, directly optimizing the zero-offset to meet measurement requirements. The redundant design of the symmetric pressure-guiding channels reduces the risk of single-point failures, ensuring functional safety in harsh environments.

In addition to structural innovations, the proposed TGV glass packaging (Table 2) offers several key advantages over traditional packaging. The reduced package size 2×2 mm and thickness ≤1.1 mm facilitate high-density SiP integration. This supports combined sensing and processing functions, in contrast to the discrete-sensing-only capability of traditional designs. Besides, it has superior compatibility with corrosive refrigerants and maintains resistance to conductive media and organic media, ensuring reliability in extreme operating conditions. It also features a higher via aspect ratio 10:1, a thinner substrate (≥135 μm), and improved RDL uniformity (10%).

Table 2. Comparison between Proposed TGV Glass Packaging and Traditional Packaging

Item			Traditional Packaging		Proposed Packaging
Basic Specifications	Package Size/mm		>Φ15		2×2
	Package Thickness/mm		>3		1.1
	Integration Level		Discrete		SiP
	Function		Sensing		Sensing+processing
Protection	Conductive Medium	Water	√	Waterproof Adhesive	√
		NaCl Solution	√		√
	Organic Medium	Oil	√		√
		N-Hexane	√		√
		H_2S	√		√
	Corrosive Refrigerant Medium	R744	×	Ceramic	√
		R1270	×		√
		R1234yf	×		√
		R134a	√		√
		R152a	×		√
		R410	×		√
Substrate Specifications	Via Aspect Ratio		5:1		7~10:1
	Substrate Thickness		≥500μm		≥135μm
	RDLUniformity		±20%		10%
Mechanical Performance	Overload Pressure		>3x		←
	Burst Pressure		>5x		←

6. Conclusion

This study develops a wafer-level glass-silicon heterogeneous integrated SiP for MEMS differential pressure sensors, addressing the drawbacks of conventional packaging. The circular bonding cavity improves zero-offset with guiding channels effectively enhancing functional safety. The TGV glass packaging facilitates high-density integration, demonstrates high environmental adaptability, and improves electrical performance. The sensor exhibits favorable core performance metrics and, when integrated with an ASIC, shows enhanced electrical robustness and mechanical durability compared to traditional designs, providing a viable approach for advanced miniaturized pressure sensing applications.

7. References

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